Chapter 4, Section 3

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1. If X is a curve of genus 2, show that a divisor D is very ample $\iff \deg D \geq 5$. This strengthens (3.3.4).

Proof. Let D be a very ample divisor on X. By the degree-genus formula, a nonsingular plane curve cannot have genus equal to 2, which implies $\dim |D| \geq 3$. Applying the Riemann-Roch theorem to D gives

$$\deg D = \dim |D| - \dim |K - D| + 1 \geq 4 - \dim |K - D|.$$

By (Ex. 1.5), D has degree at least 3 (Ex. 1.5), so $K - D$ has negative degree. Hence, $|K - D|$ has dimension -1. □

3. If X is a curve of genus ≥ 2 which is a complete intersection (II, Ex. 8.4) in some $\mathbb{P}^n$, show that the canonical divisor K is very ample. Conclude that a curve of genus 2 can never be a complete intersection in any $\mathbb{P}^n$.

Proof. Let d be the degree of X in $\mathbb{P}^n$, and assume X is not contained in any hyperplane. Then X is a complete intersection of $n-1$ hypersurfaces of degree d_i, and by (II, Ex. 8.4), we have $\mathcal{O}_X(K) \cong \mathcal{O}_X(e)$, where $e = \sum d_i - n - 1$, so it is enough to show $e \geq 1$. We have two ways of computing $\deg K$, which are $\deg K = ed$ and $\deg K = 2g - 2 \geq 2$. Hence, $e \geq 1$. The linear system given by the canonical sheaf of a curve of genus 2 is the unique g^1_2 (Ex. 1.7), which is never ample. □

6. *Curves of Degree 4.*

(a) If X is a curve of degree 4 in some $\mathbb{P}^n$, show that either
   (1) $g = 0$, in which case X is either the rational normal quartic in $\mathbb{P}^4$ (Ex. 3.4) or the rational quartic curve in $\mathbb{P}^3$ (II, 7.8.6), or
   (2) $X \subseteq \mathbb{P}^2$, in which case $g = 3$, or
   (3) $X \subseteq \mathbb{P}^3$ and $g = 1$.

(b) In the case $g = 1$, show that X is a complete intersection of two irreducible quadric surfaces in $\mathbb{P}^3$ (I, Ex. 5.1).

Proof.

(a) $\boxed{n = 2}$ The degree-genus formula gives

$$g = \frac{(d-1)(d-2)}{2} = 3.$$

$\boxed{n = 3}$ If X is not contained in any hyperplane, then $g < 3$ (III, Ex. 9.11) (Ex. 3.5b). Since $g \neq 2$ by (Ex. 3.1), X is either the rational quartic curve in $\mathbb{P}^3$, or an elliptic curve.

$\boxed{n = 4}$ X is rational by (Ex. 1.5).

(b) Let $\mathcal{I}$ be the ideal sheaf of X in $\mathbb{P}^3$, and consider the exact sequence

$$0 \longrightarrow \mathcal{I} \longrightarrow \mathcal{O}_{\mathbb{P}^3} \longrightarrow \mathcal{O}_X \longrightarrow 0.$$

Twisting by $\mathcal{O}_{\mathbb{P}_3}(2)$ and taking cohomology gives

$$0 \longrightarrow \mathrm{H}^0(\mathbb{P}^3, \mathcal{I}(2)) \longrightarrow \mathrm{H}^0(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}(2)) \longrightarrow \mathrm{H}^0(X, \mathcal{O}_X(2)) \longrightarrow \cdots.$$

Now, X is an elliptic curve, and $\mathcal{O}_X(2)$ corresponds to a divisor D of degree 8. The Riemann Roch theorem gives $\dim \mathrm{H}^0(X, \mathcal{O}_X(2)) = 8$, and the vector space in the middle has dimension 10, so we conclude that $h^0(\mathbb{P}^3, \mathcal{I}(2)) \geq 2$. An element of $\mathrm{H}^0(\mathbb{P}^3, \mathcal{I}(2))$ is a form of degree 2, whose zero-set is a quadric surface containing X. Any such quadric must be irreducible and reduced because X is not contained in any hyperplane. Hence, X is contained in two distinct irreducible quadric surfaces and is a complete intersection because X is a curve of degree 4.

□

9. Prove the following lemma of Bertini: if X is a curve of degree d in $\mathbb{P}^3$, not contained in any plane, then for almost all planes $H \subseteq \mathbb{P}^3$ (meaning a Zariski open subset of the dual projective space $(\mathbb{P}^3)^*$), the intersection $X \cap H$ consists of exactly d distinct points, no three of which are collinear.

Proof. Bertini's theorem says a general plane in $\mathbb{P}^3$ intersects X in d distinct points. To show that no three of these points are collinear generically, it is enough to show a general plane in $\mathbb{P}^3$ does not contain a multisecant of X. The proof of (3.10) shows that the set of multisecants of X has dimension ≤ 1 in $(\mathbb{P}^3)^*$, and a general plane in $\mathbb{P}^3$ corresponds to a line in $(\mathbb{P}^3)^*$. Hence, for almost almost all planes $H \subseteq \mathbb{P}^3$, the intersection $X \cap H$ consists of exactly d distinct points, no three of which are collinear. □

12. For each value of $d = 2, 3, 4, 5$ and r satisfying $0 \leq r \leq \frac{1}{2}(d-1)(d-2)$, show that there exists an irreducible plane curve of degree d with r nodes and no other singularities.

Proof. There is nothing to prove for $d = 2$, so fix any $d \geq 3$, and denote $g = \frac{1}{2}(d-1)(d-2)$.

$$d = 2, 3, 4, 5, \quad g = 0, 1, 3, 6$$

Both extremes $r = 0, g$ are consructed in (3.11.1).

If X is any elliptic curve and P is any point in X, then the linear system $|dP|$ gives an embedding of X in $\mathbb{P}^{d-1}$ as a curve of degree d. Projecting the curve into $\mathbb{P}^2$ by (3.5) and (3.10) gives a plane curve of degree d and $g-1$ nodes. This gives $d = 4$ and $r = 2$.

By (Ex 3.6), the complete intersection of two quadric surfaces in $\mathbb{P}^3$ is an elliptic curve of degree 4. Projecting from a point not in the curve by (3.10) gives a plane curve of degree $d = 4$ and $r = 2$ nodes

Similarly, any divisor of degree ≥ 5 on a genus 2 curve is very ample by (Ex. 1.5), so there exists plane curves of degree $d \geq 5$ with $g - 2$ nodes. This gives $d = 5, r = 4$.

If $Q \subset \mathbb{P}^3$ is the nonsingular quadric suface, then there exists divisors of type (a, b) for any $a, b > 0$ that are linearly equivalent to an irreducible nonsingular curve of degree $a + b$ and genus $(a-1)(b-1)$ as a curve in $\mathbb{P}^3$.

A divisor of type $(2, 3)$ gives a nonsingular curve of degree $2 + 3 = 5$ and genus $(2-1)(3-1) = 2$ in $\mathbb{P}^3$. Projecting into $\mathbb{P}^2$ by (3.10) gives a plane curve of degree $d = 5$ and $r = 4$ nodes. Projecting this curve from one of its own points that does not lie in a plane spanned by coplanar tangent lines gives a plane curve of degree $d = 4$ and $r = 1$ node.

Now, consider a divisor of type $(3, 3)$. It is a nonsingular curve of degree $3+3 = 6$ and genus $(3-1)(3-1) = 4$ in $\mathbb{P}^3$. Projecting this curve from one of its points that does not lie in a plane spanned by coplanar tangent lines gives a plane curve of degree $d = 5$ and $r = 2$ nodes.

Finally, consider the divisor of type $(4, 2)$. It has degree $2 + 4 = 6$ and genus $(4-1)(2-1) = 3$ as a curve in $\mathbb{P}^3$. Projecting this curve from one of its points that does not lie in a plane spanned by coplanar tangent lines gives a plane curve of degree $d = 5$ and $r = 3$ nodes.

More generally, any divisor of degree $d \geq 2g + 1$ on any curve of genus g is very ample by (3.2), so there exists plane curves of degree d and $g - d$ nodes. This gives $d = 5$ and $r = 1$.

□